

Challenge Write-Up Template

Challenge Information

Student Name	Evallo, Daniella Kim N.
Section / Class	BSIT-3G
Challenge Name	Multicode
Category	General Skills
Points / Difficulty	Easy
Date Solved	08/09/2026
Flag Format Given	PicoCTF{...}

1. Challenge Description

The challenge provided a suspiciously encoded message and explained that the flag was hidden under several layers of common encodings. The task was to download the message, identify the encoding layers, reverse them in the correct order, and recover the flag.

2. Initial Analysis / Reconnaissance

I first downloaded the message file and opened it to see what was inside. The contents were not readable text; instead, it was a long encoded string. The challenge hints then pointed to common encodings such as ROT13, URL encoding, Hex, and Base64, which indicated that this was an encoding/decoding challenge rather than encryption.

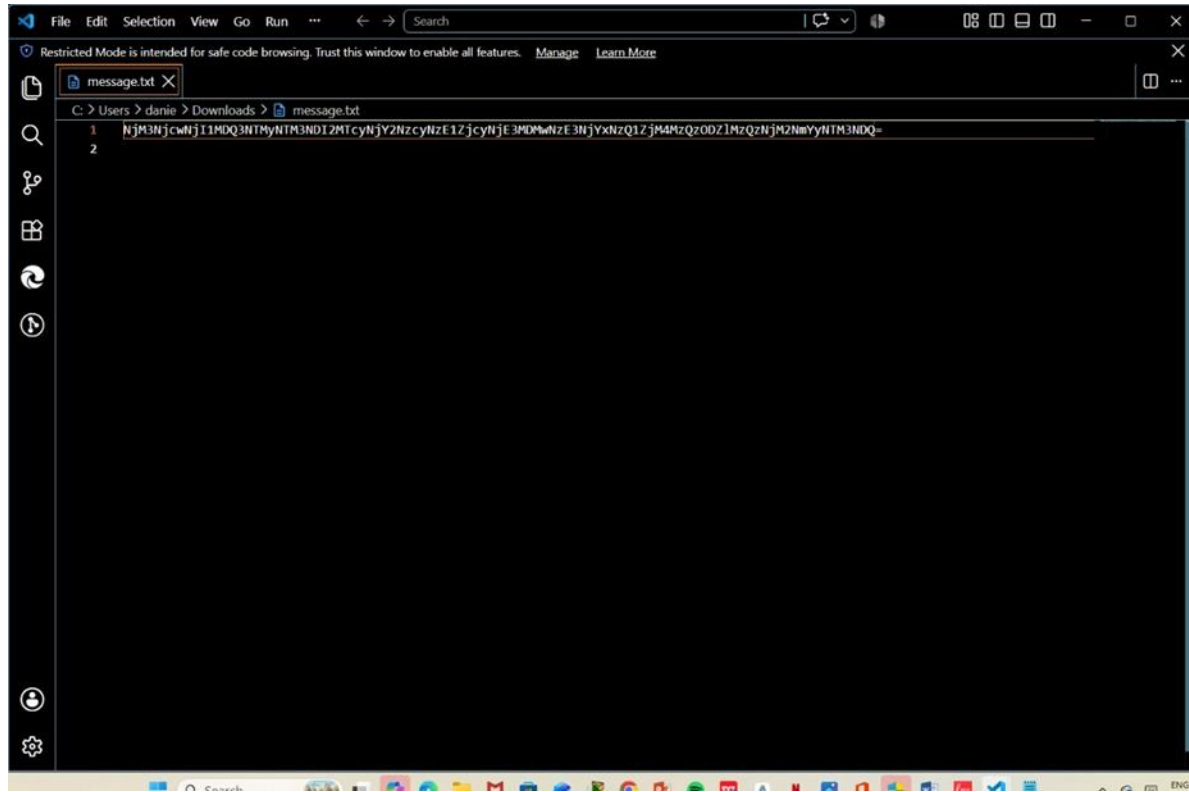
3. Tools and Commands Used

Tool / Command	What it was used for	Result / Output
message.txt	To inspect the downloaded challenge file and identify the encoded message.	A long encoded string was provided.
CyberChef	To apply the decoding operations in sequence and peel back the encoding layers.	The message decoded through Base64, Hex, URL Decode, and ROT13.
Base64 / Hex / URL Decode / ROT13	To reverse the layers indicated by the challenge hint.	The final output revealed the picoCTF flag.

4. Solution Walkthrough

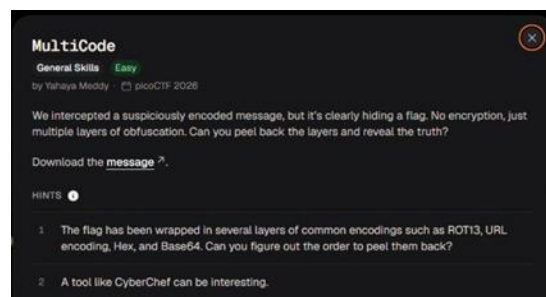
Step 1

I downloaded the challenge message and opened the file in a text editor. The file contained a long string instead of a normal sentence, so I treated it as encoded data that needed to be decoded layer by layer.



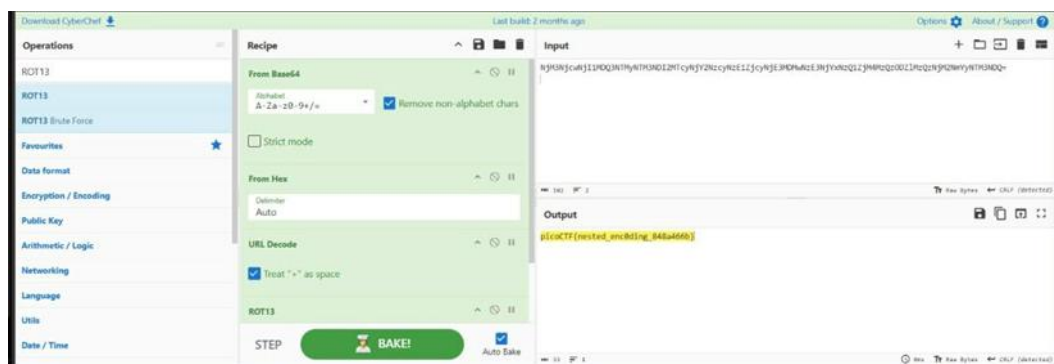
Step 2

I checked the challenge hints before decoding. The first hint identified the likely layers as **ROT13, URL encoding, Hex, and Base64**. This gave me the operations to test in CyberChef and helped determine that the message had been wrapped in several encodings.



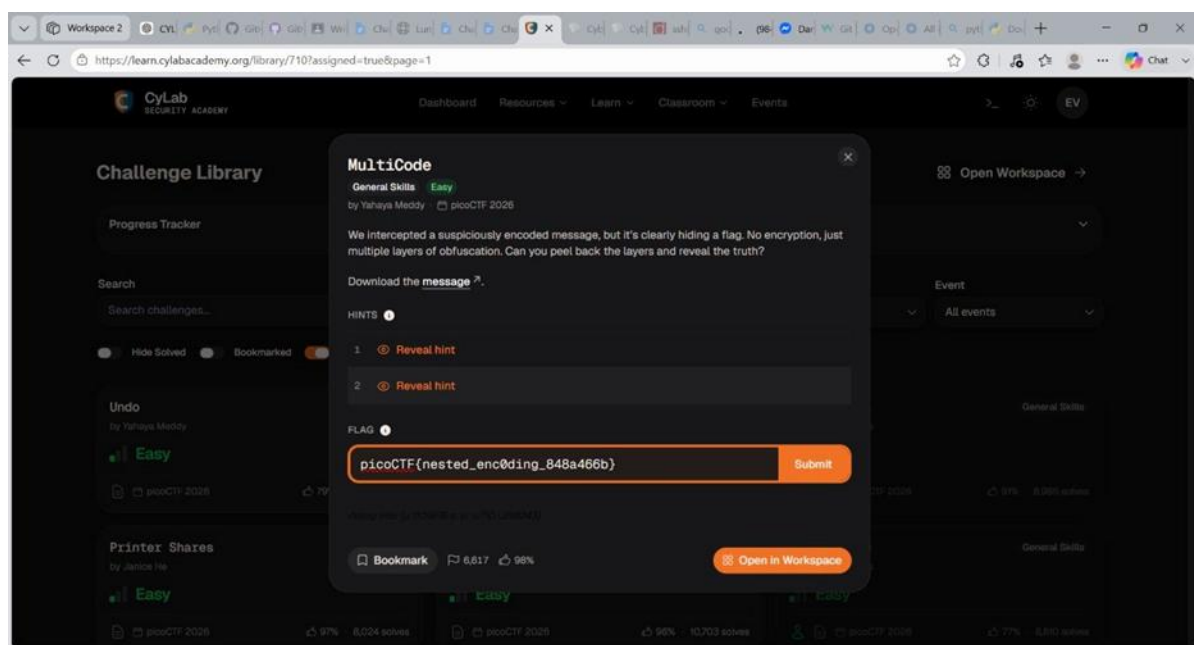
Step 3

I used **CyberChef** to reverse the layers. The recipe shown in the evidence uses **From Base64, From Hex, URL Decode, and ROT13**. After applying these operations in sequence, the output changed from the encoded text into the readable flag.



Step 4

Then submit the recovered flag in the challenge page. The submission field accepted the value, confirming that the decoding sequence was correct.



5. Flag

picoCTF{nested_enc0ding_848a466b}

6. Key Concept / What I Learned

The main concept in this challenge was recognizing the difference between encryption and encoding. The message used multiple common encoding layers, so the solution was to identify each layer and reverse them in the correct order. CyberChef made it easier to test and chain the required decoding operations.

7. References